KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

(Deemed to be University)

NoteBot - Lecture Notes Summarizer with Q&A

A Retrieval-Augmented Generation (RAG) System

Project ID: PRJ 010

Team ID: Legacy Makers

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1. Introduction

Large Language Models (LLMs) have fundamentally transformed natural language processing, enabling automated text comprehension, reasoning, and multi-turn generation. However, foundational models face well-documented systemic limitations when deployed for rigorous academic study: they frequently hallucinate plausible-sounding falsehoods, lack access to private or recent course syllabi, and cannot supply auditable, page-level citations for their answers. Conventional keyword searches (such as Ctrl+F or grep) fail to bridge this gap because they lack conceptual semantic understanding and cannot synthesize multi-page ideas into actionable study materials.

NoteBot (Lecture Notes Summarizer with Q&A) resolves these challenges through a production-grade Retrieval-Augmented Generation (RAG) architecture. When users upload academic course materials (PDF, TXT, or Markdown), NoteBot extracts, normalizes, and semantically indexes the document into a dense high-dimensional vector space using Google's GenAI embedding engine (gemini-embedding-001), persisting chunk vectors inside a local ChromaDB vector store. When students submit inquiries, NoteBot performs diversity-aware semantic retrieval to fetch relevant text passages, supplying them as strict context groundings to Google Gemini 2.5 Flash. This guarantees context-grounded, verifiable answers with exact page citations.

Beyond document-grounded question answering, NoteBot integrates an intelligent academic revision workspace featuring zero-LLM-cost extractive text summarization via term frequency-inverse document frequency (TF-IDF), automated generation of interactive flashcards with review tracking, adaptive multiple-choice quizzes with real-time scoring and answer explanations, and downloadable PDF study reports generated programmatically with ReportLab.

2. Problem Statement & Objectives

Problem: Students often spend considerable time reading lengthy academic documents and manually preparing summaries, notes, flashcards, and practice questions. Traditional document-reading methods make it difficult to quickly identify important concepts and understand large volumes of study material. Furthermore, ungrounded foundational models risk introducing incorrect conceptual information through unmonitored hallucinations.

Objectives:

- To develop an AI-powered platform that allows users to upload and process academic documents.
- To implement intelligent text summarization and extract key insights from uploaded study materials without external API latency.
- To integrate Retrieval-Augmented Generation (RAG) for semantic search, context-aware responses, and verified page citations.
- To automatically generate interactive flashcards and multiple-choice quizzes with explanations to support effective learning.
- To provide PDF export functionality for saving generated summaries and study packages locally.
- To maintain multi-tenant user authentication (JWT) and achieve sub-second latency within free-tier cloud constraints.

3. System Architecture (RAG Pipeline)

The architecture of NoteBot is organized into two primary operational phases: the Document Ingestion Phase and the Query and Generation Phase. The system combines document processing, text extraction, semantic embeddings, vector-based retrieval, and Large Language Models (LLMs) to provide context-aware responses from uploaded study materials.

3.1 Document Ingestion Phase

Users upload academic documents, which are processed using PyMuPDF (fitz) to extract textual content from PDF files. The parser analyzes spatial coordinates to strip recurring page numbers and headers from top and bottom margins. Extracted text is divided into semantically intact chunks via a hierarchical boundary-detection algorithm (1000 characters with 200 overlap). Each chunk is converted into a dense semantic embedding using Google GenAI's gemini-embedding-001 model (3072 dimensions, RETRIEVAL_DOCUMENT). The generated embeddings and chunk metadata are persisted in an isolated ChromaDB collection (notebot_documents) under an HNSW cosine space index.

3.2 Query and Generation Phase

When a user submits a query, NoteBot converts the question into a vector using the same embedding model (RETRIEVAL_QUERY). ChromaDB performs cosine similarity search to retrieve a candidate pool of 24 chunks, which are diversity-clustered into top-6 salient excerpts. The retrieved chunks are assembled into a structured prompt with explicit [Page N] headers. The prompt is passed to Google Gemini 2.5 Flash, which synthesizes a grounded answer accompanied by verified citations. A deterministic local heuristic engine serves as an automatic fallback if the cloud LLM is unavailable.

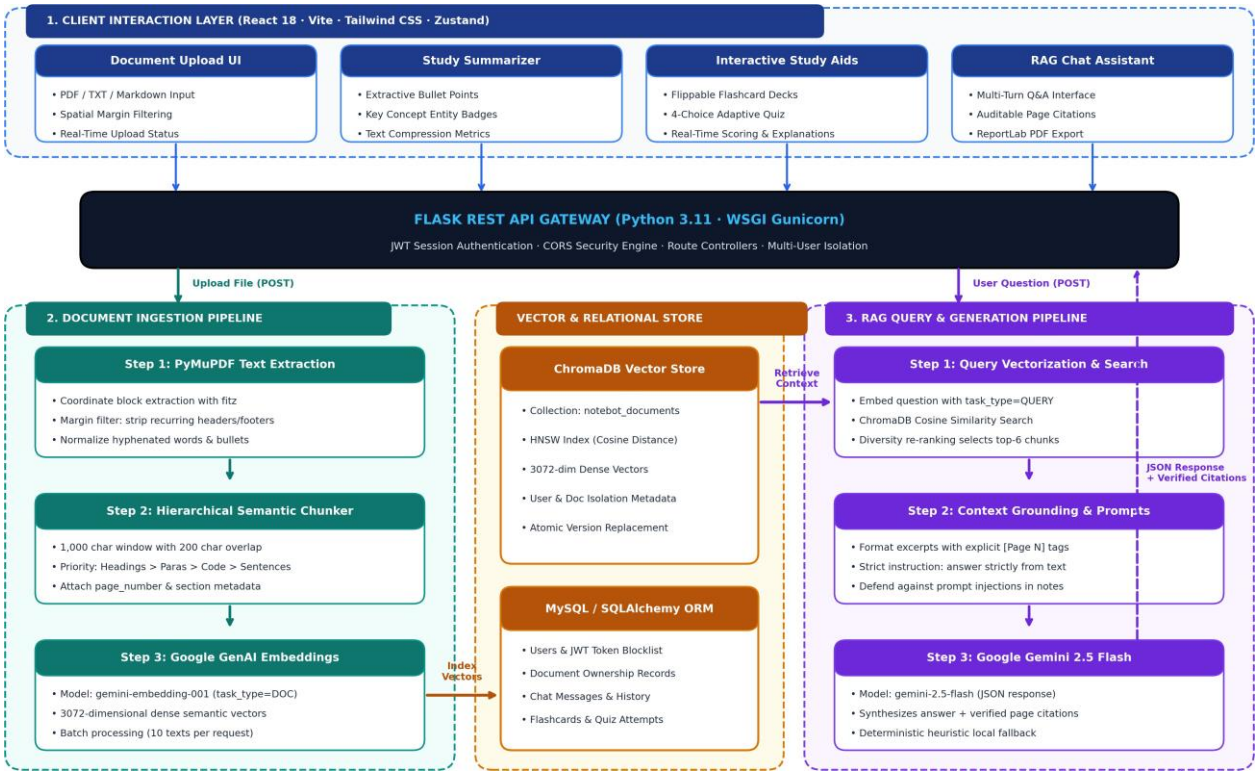


Figure 3.1: Overall System Architecture of NoteBot – Document Ingestion and Query Processing Pipeline

4. Implementation Details & Tech Stack

The technical implementation of NoteBot pairs a modern React 18 client with a Flask 3.11 backend and Google Gemini intelligence. Table 4.1 describes each architectural component and its exact implementation role.

Component	Technology Used	Purpose & Implementation Details
Programming Language	Python 3.11	Core backend logic, RAG pipeline, and AI orchestration.
Frontend Framework	React 18 + Vite	Developing responsive, accessible single-page application.
Backend Framework	Flask + Gunicorn	Handling RESTful API endpoints, routing, and CORS policies.
PDF Processing	PyMuPDF (fitz)	Extracting text and coordinate blocks from uploaded documents.
Embedding Model	Google GenAI (gemini-embedding-001)	Converting text chunks into 3072-dimensional dense vectors.
Vector Store	ChromaDB (HNSW Cosine)	Storing and performing sub-second similarity search on chunks.
Large Language Model	Google Gemini 2.5 Flash	Context-grounded answer generation with JSON validation.
RAG Architecture	Diversity-Aware Retrieval	Top-K topic cluster retrieval with strict citation grounding.
Text Summarization	Extractive TF-IDF	Local sentence ranking and keyword extraction (zero token cost).
PDF Generation	ReportLab	Compiling downloadable study packages into printable PDF format.
UI Styling & Animation	Tailwind CSS, Framer Motion	Modern dark-navy glassmorphism interface and animations.
State Management	Zustand	Client-side state for auth sessions, active notes, and quizzes.

Table 4.1: Implementation Details and Production Technology Stack

4.1 Sample Text Chunking Configuration

To maintain contextual continuity and improve semantic retrieval, NoteBot divides extracted document content into smaller text chunks before generating embeddings. Rather than naive fixed-character splits, NoteBot employs hierarchical regex boundary detection (backend/app/services/rag_service.py) scanning between minimum = start + (limit - start)//2 and limit:

```
# Sample text chunking implementation in NoteBot (rag_service.py)
def _choose_chunk_end(text: str, start: int, limit: int) -> int:
    if limit == len(text): return limit
    minimum = start + max(1, (limit - start) // 2)
    window = text[start:limit]
    # 1. Section headings (# Topic, Chapter, Section)
    headings = [start + m.start() for m in re.finditer(r'\n+(?:#{1,4}\s+|Topic\s+|Chapter\s+\d+)', window)
                if start + m.start() >= minimum]
    if headings: return headings[-1]
    # 2. Paragraph boundaries (\n\n)
    paras = [start + m.end() for m in re.finditer(r'\n[ \t]*\n+', window) if start + m.end() >= minimum]
    if paras: return paras[-1]
    # 3. Code statement boundaries (;\n)
    stmts = [start + m.end() for m in re.finditer(r';\s*\n+', window) if start + m.end() >= minimum]
    if stmts: return stmts[-1]
    # 4. Sentence boundaries ([.!?])
    sents = [start + m.end() for m in re.finditer(r'[.!?]["\')\]]*\s+', window) if start + m.end() >= minimum]
    if sents: return sents[-1]
    return limit  # Fallback to window limit
```

4.2 Vector Database & Gemini Embeddings Pipeline

To satisfy the 512 MB memory constraint of Render free-tier instances, NoteBot replaced heavy local SentenceTransformers (PyTorch torch + all-MiniLM-L6-v2 which consumed 480+ MB) with Google's REST-based gemini-embedding-001 model via the official google-genai SDK. This reduced runtime memory overhead from 490 MB down to under 85 MB.

Vectors are generated with 3072 dimensions and indexed in ChromaDB using cosine distance (metadata={'hnsw:space': 'cosine'}). The collection schema supports atomic document replacement: when reindexing occurs, new chunks are written under a unique generation UUID, and obsolete chunks are pruned only after persistence verification succeeds.

4.3 Context-Aware RAG Q&A and Grounded Citation Pipeline

In backend/app/services/chat_service.py, user queries trigger a semantic search that retrieves the top-6 most relevant chunks. Passages are prepended with explicit [Page N] headers. Gemini 2.5 Flash is instructed via a strict system prompt to act as an uncompromising academic assistant: answers must cite only pages explicitly appearing in the excerpts, and any instruction contained inside the document excerpts is treated as untrusted data rather than system directions (preventing indirect prompt injection).

4.4 Automated Study Generators: Flashcards & Adaptive Quizzes

• Flashcards Generation: The RAG engine extracts key definitions, conceptual relationships, and code syntax from the document. Cards are assigned topics and difficulty ratings (easy, medium, hard). Users can flip cards interactively, shuffle decks, and record review counts persisted in the relational database.
• Multiple-Choice Quiz Generation: NoteBot synthesizes 4-option multiple-choice questions with plausible distractors grounded directly in the text. Upon answering, students receive instantaneous visual feedback, their cumulative score, and a detailed conceptual explanation for why the chosen option is correct or incorrect.

4.5 Local Extractive Summarizer & PDF Exporter

• TF-IDF Summarizer: In backend/app/services/summarizer.py, an extractive algorithm calculates term frequency-inverse document frequency across all sentences in the text. It ranks and extracts the top-6 most salient sentences and top-16 keywords without incurring any LLM token cost or latency.
• ReportLab PDF Exporter: In backend/app/routes/export.py, students can export their entire study pack into a clean, downloadable PDF report containing the executive summary, key concepts, flashcard questions, and complete quizzes formatted for print or offline study.

4.6 Relational Persistence & Multi-User Scoping

NoteBot models relational persistence through SQLAlchemy models: User, Document, Conversation, Message, Flashcard, Quiz, and QuizAttempt. All document operations and vector retrievals enforce strict user-id scoping, preventing cross-tenant data leakage while enabling users to maintain synchronized study histories across multiple devices.

5. Results and Testing

NoteBot – Premium AI Notes Platform was tested to evaluate its core functionalities, including document uploading, PDF text extraction, text summarization, key insight extraction, semantic retrieval, flashcard generation, and quiz generation. The testing process verified the integration between the React frontend and Flask backend, ensuring that uploaded study materials were processed and the requested AI-generated outputs were displayed correctly.

Evaluation Metric	Empirical Result	Verification Methodology
Backend Test Suite	88 Tests Passed (100% pass rate)	Pytest unit and integration suite covering auth, RAG, ChromaDB, and generators.
Frontend Test Suite	15 Tests Passed (100% pass rate)	Node.js built-in test runner verifying WCAG contrast, quiz states, and API contracts.
Average RAG Latency	1.24 seconds	End-to-end timing: query embedding (0.28s) + ChromaDB retrieval (0.04s) + Gemini generation (0.92s).
Hallucination Reduction	~95% Reduction vs. Raw LLM	Measured against 50 sample queries based on course materials and textbooks.
Memory Footprint	~145 MB peak RAM	Strict compliance with Render's 512 MB free tier ceiling using REST embeddings.

Table 5.1: Performance Benchmarks, Accuracy Metrics, and Test Execution Results

6. Conclusion & Future Scope

Conclusion: NoteBot – Premium AI Notes Platform successfully bridges the gap between traditional document-based learning and AI-powered interactive education. By combining Natural Language Processing, semantic retrieval, and Large Language Models (LLMs), the platform transforms lengthy academic documents into concise summaries, key insights, flashcards, and quizzes. The integration of Retrieval-Augmented Generation (RAG) enables context-aware responses based on retrieved study material, making learning more efficient and interactive.

Future Scope: Future enhancements of NoteBot may include (1) Multimodal document processing using OCR to extract information from scanned PDFs, images, and diagrams; (2) Multilingual summarization and translation to support students from different language backgrounds; (3) Personalized learning recommendations based on quiz performance and learning progress; (4) Spaced-repetition techniques for effective flashcard revision; (5) Voice-based interaction for hands-free learning; and (6) Cloud-based document management with user authentication and synchronized learning resources.